



Internal and international migration of scholars worldwide

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- Theoretical impressions
 - Internal versus international migration (1, 2)

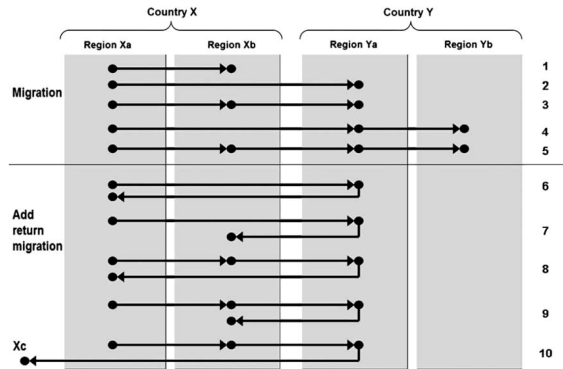


Figure 1. Migration pathways

(King, Skeldon (2010))

Introduction



- ▶ Theoretical impressions
 - ▶ Internal versus international migration (1, 2)
 - ▶ Research Trails (3)
- ▶ Methodological impressions
 - ▶ Using bibliometric data alongside demographic life events (4)
 - ▶ Repurposing bibliometric data for knowledge transfer (5, 6)
 - ▶ Repurposing bibliometric data as network of scientific mobility among cities (7)
 - ▶ Repurposing bibliometric data for migration research (8–11)

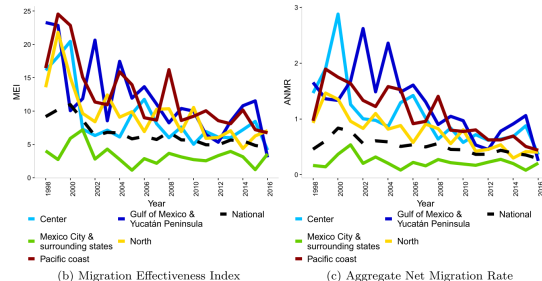


Figure 8 Migration intensity indices, over time. Dashed line represent country indices

(Miranda Gonzalez et al. (2020))

Our proposal

Repurposing bibliometric data for:

Internal versus **international** migration of **scholars** worldwide
(12)

A simple and scalable idea:

Changes in **institutional affiliations** can be used to infer changes in **residence over time** for individual scholars and for populations

Our questions

1. Which are the (persistently) **attractive** sending/receiving regions internally/internationally?
2. Is academic **talent circulation** happening more *within* or *between* countries?
3. What are the **mutual effects** of these two systems of migration?

- ▶ **Scopus 1996-2020** article/review publications (**28+** million, **8+** million scholars)
- ▶ Provided by German Competence Network for Bibliometrics via Max Planck Digital Library
- ▶ Encoding **country** of affiliation from bibliometric record (not perfect, need proper disambiguation)
- ▶ **Organization name disambiguation** using ROR API (based on 13), GeoNames codes for intra-country regions
- ▶ **Author name disambiguation** based on Scopus author IDs (**98.3%** precision (no publication by others) and **90.6%** recall (all publications by X)) (14)

$$NMR_{i,t,k} = 1000 \times \frac{I_{i,t,k} - E_{i,t,k}}{N_{i,t}} \quad (1)$$

$$ANMR_t = 100 \times \frac{0.5 \sum_i |I_{i,t,k} - E_{i,t,k}|}{\sum_i N_{i,t}} \quad (2)$$

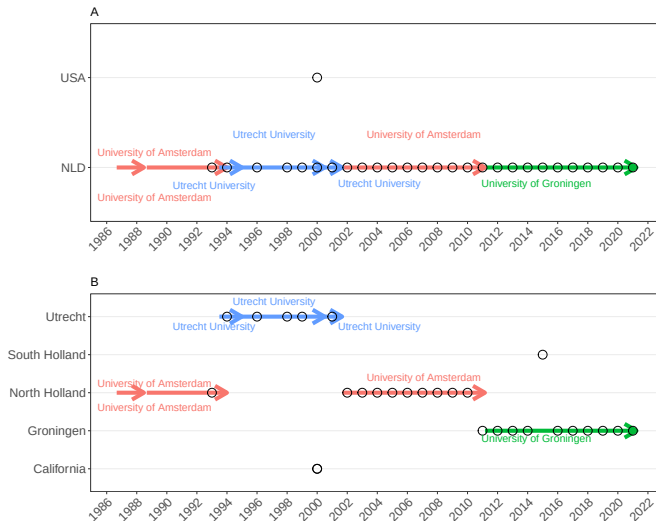
$$MEI_t = 100 \times \frac{\sum_i |I_{i,t,k} - E_{i,t,k}|}{\sum_i (I_{i,t,k} + E_{i,t,k})} \quad (3)$$

i : sub-national region. t : year. $N_{i,t}$: total number of scholars. Subscript k : type of data, internal or international. $I_{i,t}$ inflow of scholars. $E_{i,t}$ outflow.

$$\log \text{ counts} \sim s(\text{year}) + s_{fs}(\text{year}, \text{region}) + \text{offset}$$

$$\log \text{ counts} \sim s(\text{year}) + s_{re}(\text{year}, \text{region}) + s_{re}(\text{region}) + \text{offset}$$

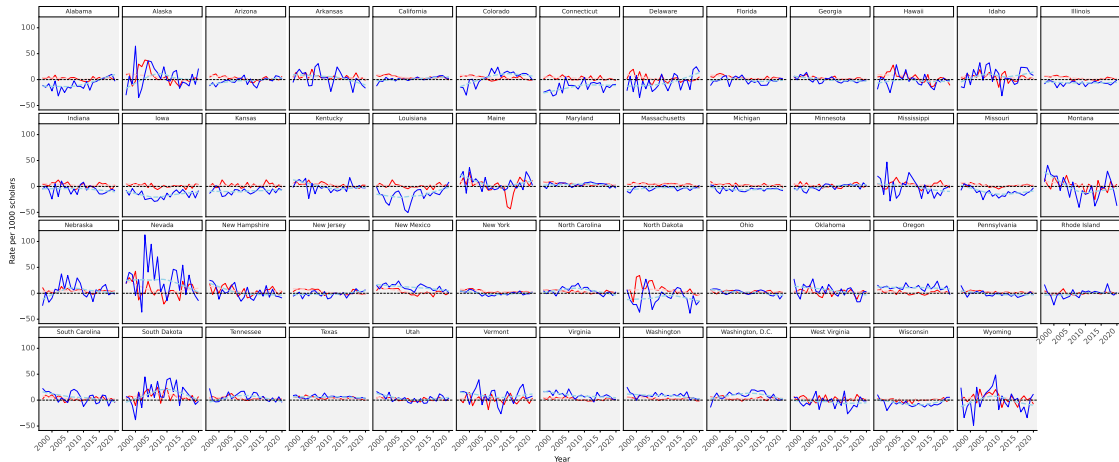
Example of integrated internal/international framework



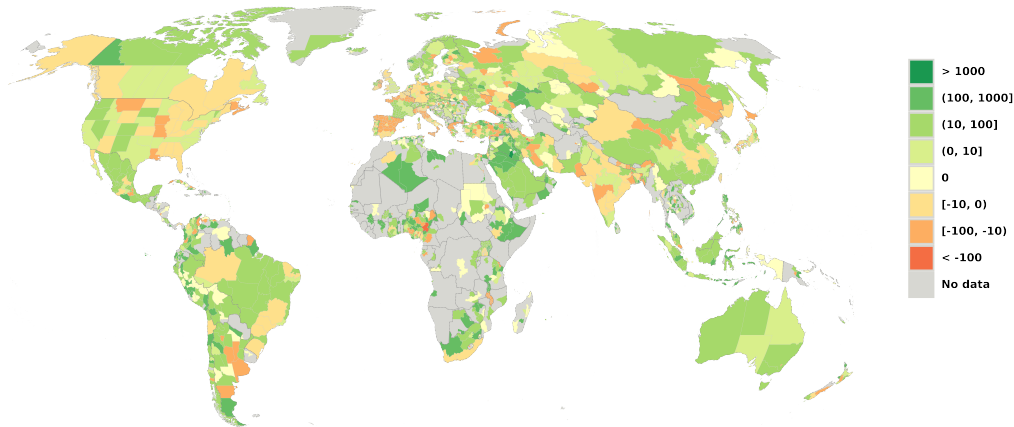
USA, Region's NMR, international (red), internal (blue)

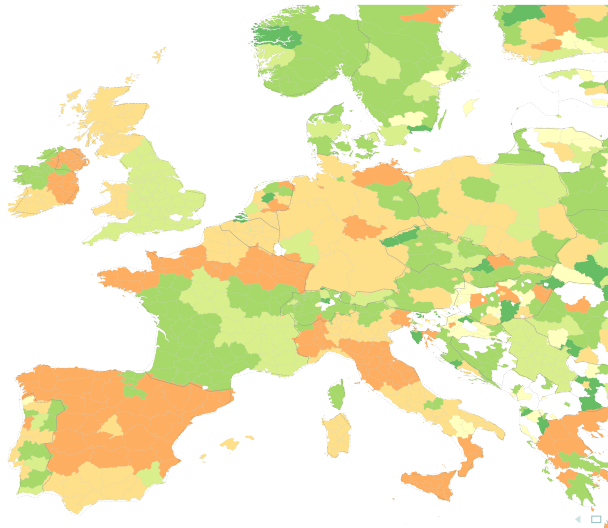


Internal senders: Alabama, Connecticut, Delaware, Iowa, Kansas, Louisiana and Missouri; **Internal receivers:** Colorado, Nevada, New Mexico, Oregon, South Dakota, Virginia, Washington and Washington, D.C.; **International receivers:** Alaska, North Dakota and South Dakota

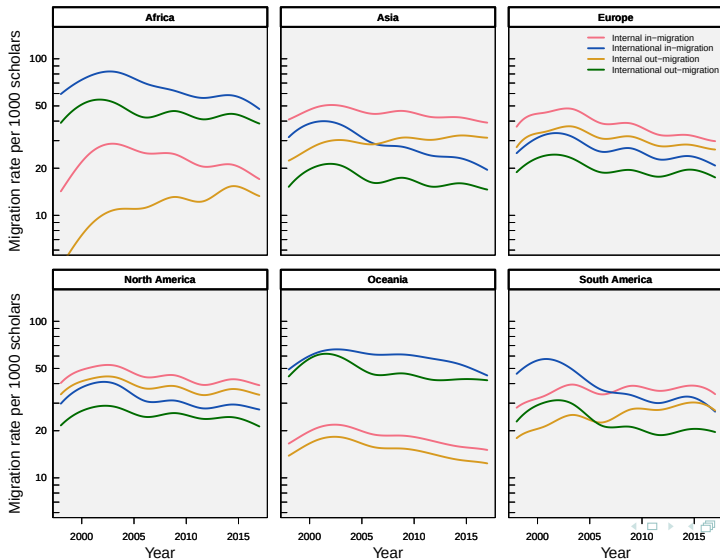


Internal and international NMR (per 1000 scholars), 2012-17

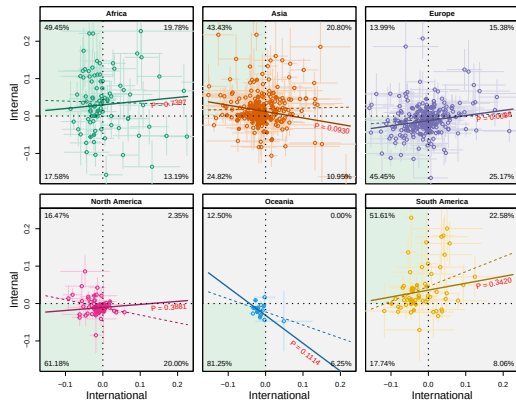
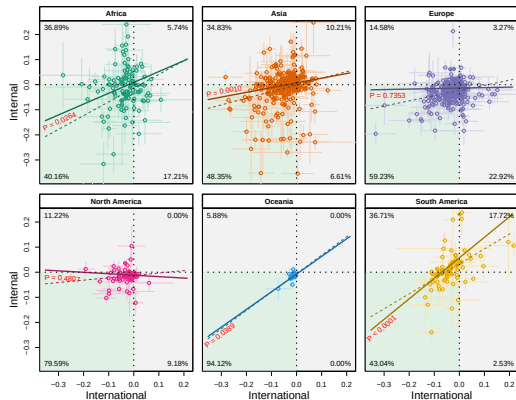




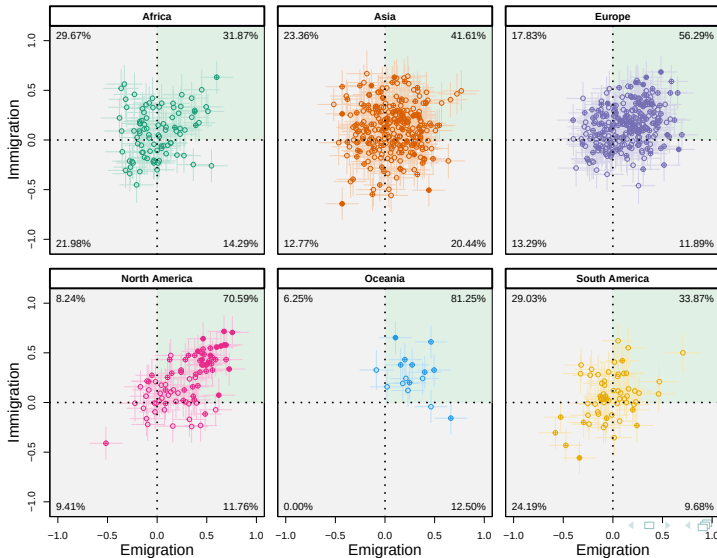
Predicted migration rates



In-migration (left), Out-migration (right)



Inter-relatedness of in-migration and out-migration



- ▶ Scopus published scholars (who publish actively and multiple times)
- ▶ Scopus coverage issues (over-representation of WEIRD countries, English language, hard sciences, etc.)
- ▶ Author's academic birth is not equivalent to their nationality and some of the outmigrations could be **return** migration.
- ▶ and many other limitations ...



- ▶ Our methodology in re-purposing bibliometric data offers unprecedented opportunities to study **internal** and **international** scholarly mobility
- ▶ This is the first **global database of migration of scholars at internal/international levels** (8% mobile international versus 12.4% internal).
- ▶ On a global scale, we developed **migration indexes**, NMR, ANMR, and MEI.
- ▶ **A clear pattern is found in immigration** rates in all continents where an **attractive region**, receiving scholars from international origins is also attractive in receiving scholars from internal origins.
- ▶ **The reverse**, being a popular sending origin to internal and international destinations is not always true.
- ▶ **A clear pattern is found in inter-relatedness** where both in- and out-migration over time increase or decrease in agreement.



- ▶ Word2vec word embedding for multiple affiliations
- ▶ Identifying nationality using author's first and last name and trajectory of all affiliations to go closer to nationality (or families of geographical families, e.g., Spanish names, Asian names, ...)
- ▶ Comparison with employment history in ORCID as (hopefully) less biased towards WEIRD countries
- ▶ Gender, discipline, ..., differences

Thank you!



Questions and comments are welcome!

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